

GWANGJU, Korea, Republic of

WMO: 471560

Lat: 35.1731N

Lon: 126.8917E

Elev: 74

StdP: 100.44

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-5.0	-15.6	1.0	-1.0	-13.7	1.2	-0.6	7.2	2.2	6.1	0.9	2.3	50	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	33.0	25.5	31.8	24.9	30.6	24.2	26.9	30.2	26.1	29.7	25.5	29.1	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.6	28.6	25.2	20.5	28.0	24.5	19.7	27.7	85.1	30.2	81.8	30.0	79.1	29.2	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	5.8	5.1	-9.0	34.8	2.0	1.3	-10.4	35.8	-11.5	36.6	-12.6	37.3	-14.0	38.3		
			-10.2	27.2	1.7	1.3	-11.5	28.2	-12.5	28.9	-13.4	29.7	-14.7	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	1.2	3.2	7.8	13.7	19.0	23.1	26.3	22.5	16.6	9.9	3.5
		DBStd	9.40	3.38	3.71	3.69	3.46	2.59	1.90	2.44	2.37	2.60	3.00	3.57
		HDD10.0	812	273	191	88	8	0	0	0	0	1	48	203
		HDD18.3	2227	531	423	326	142	23	0	0	2	68	252	460
		CDD10.0	2471	1	2	20	120	278	392	506	522	376	206	46
		CDD18.3	844	0	0	0	4	43	143	248	264	127	16	1
		CDH23.3	6789	0	0	0	32	286	812	2256	2614	736	53	1
		CDH26.7	2065	0	0	0	1	45	182	742	940	153	2	0
(14)	Wind	WSAvg	2.0	1.9	2.0	2.1	2.2	2.2	2.1	2.5	2.2	1.9	1.6	1.7
(15)	Precipitation	PrecAvg	1337	36	43	65	100	98	180	273	255	150	56	32
(16)		PrecMax	2020	126	108	170	217	226	543	693	538	475	173	74
(17)		PrecMin	764	4	3	9	16	4	25	36	28	9	2	2
(18)		PrecStd	340	25	27	40	56	55	124	145	130	109	43	18
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	12.8	16.9	20.9	26.3	29.8	31.5	34.8	34.8	31.5	26.5	21.9	15.1
(20)		0.4% MCWB	9.3	12.6	12.2	16.4	17.8	20.6	26.0	26.1	23.4	18.4	16.3	10.7
(21)		2% DB	9.9	13.6	18.2	24.0	27.9	29.9	33.0	33.3	29.7	24.8	19.5	12.6
(22)		2% MCWB	5.8	9.0	10.8	15.2	17.2	20.5	25.7	25.7	22.5	17.2	14.2	8.4
(23)		5% DB	8.0	11.5	16.2	22.2	26.3	28.8	31.7	32.2	28.4	23.4	17.7	10.8
(24)		5% MCWB	4.5	6.9	10.3	13.9	16.8	20.4	25.2	25.4	21.8	16.6	13.1	7.4
(25)		10% DB	6.4	9.4	14.4	20.2	24.8	27.5	30.4	31.0	27.2	21.9	16.1	9.1
(26)		10% MCWB	3.3	5.2	8.8	12.9	16.4	20.2	24.9	24.9	21.2	16.1	11.9	5.8
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	10.2	14.3	15.0	19.4	22.0	24.5	27.9	27.8	25.3	22.1	17.8	12.0
(28)		0.4% MCDB	11.7	15.7	17.2	22.7	25.2	26.9	30.6	31.3	28.2	24.1	20.3	14.2
(29)		2% WB	7.0	10.2	13.1	17.2	20.1	23.5	27.0	27.0	24.5	19.9	16.0	9.5
(30)		2% MCDB	9.2	12.4	15.9	21.1	23.0	26.0	30.2	30.5	27.6	21.7	17.8	11.8
(31)		5% WB	5.2	7.4	11.3	15.7	18.9	22.7	26.2	26.3	23.7	18.3	14.2	7.7
(32)		5% MCDB	7.4	10.5	14.9	19.5	23.3	25.5	29.7	30.0	26.8	21.1	16.9	10.4
(33)		10% WB	3.5	5.8	9.6	14.4	17.9	21.9	25.6	25.8	22.7	17.0	12.6	6.2
(34)		10% MCDB	5.8	8.9	13.6	18.2	22.6	25.1	29.1	29.4	25.5	20.6	15.4	8.9
(35)	Mean Daily Temperature Range	MDBR	7.1	8.4	9.8	10.6	10.1	8.0	6.1	6.6	7.8	9.2	8.6	7.5
(36)		5% DB MCDBR	9.3	11.4	12.8	13.6	13.0	10.6	8.0	8.2	9.1	10.9	10.7	10.2
(37)		5% WB MCWBR	6.0	7.0	6.7	6.0	4.9	3.5	2.5	2.5	3.2	4.4	5.8	6.7
(38)		5% WB MCDBR	8.3	9.5	10.3	9.6	9.2	6.6	6.9	6.9	6.8	7.4	8.2	9.0
(39)	Clear-Sky Solar Irradiance	taub	0.446	0.486	0.558	0.576	0.592	0.655	0.610	0.561	0.497	0.470	0.456	0.445
(40)		taud	1.936	1.876	1.731	1.740	1.751	1.693	1.847	1.970	2.082	2.021	1.988	1.949
(41)		Ebn at Noon	709	732	715	731	726	680	708	736	764	740	695	677
(42)		Edh at Noon	146	174	219	227	228	241	205	178	152	148	138	137
(44)	All-Sky Solar Radiation	RadAvg	2.44	3.24	4.36	5.04	5.51	4.91	4.50	4.63	4.15	3.69	2.61	2.16
(45)		RadStd	0.19	0.28	0.39	0.40	0.59	0.45	0.62	0.64	0.42	0.32	0.28	0.15

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-0.88	N/A	N/A	N/A	N/A	N/A	+98	+62
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+64	N/A

Nomenclature: See separate page